

Electric Charge and Methods of Charging

Questions, Textbook; Answers, Elias Xu

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1

Consider a electrically neutral object. If this object is given a positive charge, does it mean mass increases, decrease, or stay the same? Explain.

Decreases, because a positive charge means a decrease in the number of electrons, which while they have a small mass, will nonetheless cause mass to decrease.

2

Explain why a comb that has been rubbed through your hair attracts small bits of paper, even though the paper is uncharged?
Neutral objects, when exposed to a charged object, polarize, leading to them being attracted to the charged comb.

3

A charged rod is brought near a suspended object, which is repelled by the rod. Can we conclude that the suspended object is charged?

Yes, because if it wasn't charged, the polarized object would attract to the charged rod.

4

A container holds a gas consisting of 1.5 moles of oxygen molecules. One in a million of these molecules has lost a single electron. What is the net charge of the gas?

To solve, you have to calculate the number of charged particles, and then convert those elementary charges to coulombs.

$$\frac{1.5 \text{m} \times 6.023 \times 10^{23} \frac{\text{particles}}{\text{mole}}}{1 \times 10^6 \frac{\text{particles}}{\text{charged particles}}} \times \frac{1.6 \times 10^{-19} \text{C}}{e} = \boxed{0.144 \text{C}}$$

The first part compute the number of charged particles, and the second part converts those charged particles into coulombs. $6.023 \times 10^{23} \frac{\text{particles}}{\text{mole}}$ is Avogadro's number.

5

When adhesive tape is pulled from a dispenser, the detached tape acquired a positive charge and the remaining tape in the dispenser acquires a negative charge. If the tape pulled from the dispenser has $0.14 \mu\text{C}$ of charge per centimeter, what length of tape must be pulled to transfer 1.8×10^{13} electrons to the remaining tape?

You just have to divide the amount of charge needed by the rate, after converting the electrons to coulombs.

$$1.8 \times 10^{13} \text{ electrons} \times 1.6 \times 10^{-19} \frac{\text{C}}{\text{electrons}} = 2.88 \times 10^{-6} \text{C}$$
$$\frac{2.88 \times 10^{-6} \text{C}}{0.14 \times 10^{-6} \frac{\text{C}}{\text{cm}}} = \boxed{20.57 \text{cm}}$$

6

A system of 1525 particles, each which is either a electron or a proton, has a net charge of $-5.456 \times 10^{-17} \text{ C}$ (a) How many electrons are in this system? (b) What is the mass of this system?

To solve this problem, it's better to think about how many **more** electrons there are than protons, given the negative charge. To find that number, one does

$$\frac{-5.456 \times 10^{-17} \text{ C}}{-1.6 \times 10^{-19} \frac{\text{C}}{\text{electron}}} = 341 \text{ electrons}$$

And thus, the rest of the 1184 particles are made up equally of electrons and protons, you get the answer 933 for (a)

To solve for (b), one just calculates the mass based on the predetermined number of protons and electrons.

$$\text{mass} = 592 \text{ protons} \times 1.672 \times 10^{-25} \text{ kg} + 933 \text{ electrons} \times 9.109 \times 10^{-31} \text{ kg} = 9.906 \times 10^{-25} \text{ kg}$$